



PREALG: Introduction to Algebra

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Name: _____

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Learning Objectives

- Evaluate algebraic expressions by substituting values
- Simplify expressions by combining like terms
- Solve one-step equations (addition, subtraction, multiplication)
- Translate word problems into algebraic expressions

Simplify each expression completely. Show all steps and circle your final answer.

Combining like terms

1. Combine like terms: $3x + 2x$

$$3x + 2x$$

Answer: _____

2. Simplify by combining like terms: $3x + 4x + 8 - 3$

$$3x + 4x$$

Answer: _____

3. A student writes $9x + 5x$ for the total cost of tickets. Simplify this expression.

$$9x + 5x$$

Answer: _____

4. Combine like terms: $5x + 4x$

$$5x + 4x$$

Answer: _____

5. Simplify by combining like terms: $7x + 2x + 8 - 3$

$$7x + 2x$$

Answer: _____

6. A student writes $6x + 4x$ for the total cost of tickets. Simplify this expression.

$$6x + 4x$$

Answer: _____



7. Combine like terms: $6x + 6x$

$$6x + 6x$$

Answer: _____

8. Simplify by combining like terms: $10x + 3x + 8 - 3$

$$10x + 3x$$

Answer: _____

9. A student writes $5x + 1x$ for the total cost of tickets. Simplify this expression.

$$5x + 1x$$

Answer: _____

10. Combine like terms: $2x + 3x$

$$2x + 3x$$

Answer: _____

11. Simplify by combining like terms: $4x + 4x + 8 - 3$

$$4x + 4x$$

Answer: _____

12. A student writes $8x + 5x$ for the total cost of tickets. Simplify this expression.

$$8x + 5x$$

Answer: _____

Solving one-step equations

13. Solve for x : $x + 6 = 18$

$$x + 6 = 18$$

Answer: _____

14. Solve for x : $2x = 7$

$$2x = 7$$

Answer: _____



15. After earning \$14 more, Maria has \$14 in all. Write and solve an equation to find how much she started with.

$$x + 14 = 14$$

Answer: _____

16. Solve for x: $x + 4 = 19$

$$x + 4 = 19$$

Answer: _____

17. Solve for x: $2x = 5$

$$2x = 5$$

Answer: _____

18. After earning \$20 more, Maria has \$10 in all. Write and solve an equation to find how much she started with.

$$x + 20 = 10$$

Answer: _____

19. Solve for x: $x + 9 = 19$

$$x + 9 = 19$$

Answer: _____

20. Solve for x: $5x = 4$

$$5x = 4$$

Answer: _____

21. After earning \$16 more, Maria has \$28 in all. Write and solve an equation to find how much she started with.

$$x + 16 = 28$$

Answer: _____

Evaluating expressions

22. Evaluate $6x + -2$ when $x = 5$.

$$6x - 2, x = 5$$

Answer: _____



23. A plumber charges \$24 per hour plus a \$54 service fee. What is the total cost for 4 hours? Evaluate $24x + 54$ at $x = 4$.

$$24x + 54, x = 4$$

Answer: _____

24. Evaluate $4x^2 + 3$ when $x = 3$.

$$4x^2 + 3, x = 3$$

Answer: _____

25. Evaluate $1x + 5$ when $x = 1$.

$$1x + 5, x = 1$$

Answer: _____

26. A plumber charges \$57 per hour plus a \$35 service fee. What is the total cost for 2 hours? Evaluate $57x + 35$ at $x = 2$.

$$57x + 35, x = 2$$

Answer: _____

27. Evaluate $2x^2 + -3$ when $x = 4$.

$$2x^2 + -3, x = 4$$

Answer: _____

28. Evaluate $2x + 3$ when $x = 7$.

$$2x + 3, x = 7$$

Answer: _____

29. A plumber charges \$24 per hour plus a \$24 service fee. What is the total cost for 4 hours? Evaluate $24x + 24$ at $x = 4$.

$$24x + 24, x = 4$$

Answer: _____

30. Evaluate $4x^2 + -2$ when $x = 5$.

$$4x^2 + -2, x = 5$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Combining like terms, Evaluating expressions, Solving one-step equations. All answers verified by independent computation.

Solutions



Combining like terms

1. Combine like terms: $3x + 2x$

$$3x + 2x$$

→ Both terms contain the variable x — they are like terms.

→ Add the coefficients: $3 + 2 = 5$.

→ Simplified: $5x$.

Answer: $(3 + 2)x = 5x$

2. Simplify by combining like terms: $3x + 4x + 8 - 3$

$$3x + 4x$$

→ Combine x terms: $3x + 4x = 7x$.

→ Combine constants: $8 - 3 = 5$.

→ Simplified: $7x + 5$.

Answer: $(3 + 4)x = 7x$

3. A student writes $9x + 5x$ for the total cost of tickets. Simplify this expression.

$$9x + 5x$$

→ Both terms have the same variable x .

→ Add the coefficients: $9 + 5 = 14$.

→ Simplified total cost expression: $14x$.

Answer: $(9 + 5)x = 14x$

4. Combine like terms: $5x + 4x$

$$5x + 4x$$

→ Both terms contain the variable x — they are like terms.

→ Add the coefficients: $5 + 4 = 9$.

→ Simplified: $9x$.

Answer: $(5 + 4)x = 9x$

5. Simplify by combining like terms: $7x + 2x + 8 - 3$

$$7x + 2x$$

→ Combine x terms: $7x + 2x = 9x$.

→ Combine constants: $8 - 3 = 5$.

→ Simplified: $9x + 5$.

Answer: $(7 + 2)x = 9x$

6. A student writes $6x + 4x$ for the total cost of tickets. Simplify this expression.

$$6x + 4x$$

→ Both terms have the same variable x .

→ Add the coefficients: $6 + 4 = 10$.

→ Simplified total cost expression: $10x$.

Answer: $(6 + 4)x = 10x$



7. Combine like terms: $6x + 6x$

$$6x + 6x$$

→ Both terms contain the variable x — they are like terms.

→ Add the coefficients: $6 + 6 = 12$.

→ Simplified: $12x$.

Answer: $(6 + 6)x = 12x$

8. Simplify by combining like terms: $10x + 3x + 8 - 3$

$$10x + 3x$$

→ Combine x terms: $10x + 3x = 13x$.

→ Combine constants: $8 - 3 = 5$.

→ Simplified: $13x + 5$.

Answer: $(10 + 3)x = 13x$

9. A student writes $5x + 1x$ for the total cost of tickets. Simplify this expression.

$$5x + 1x$$

→ Both terms have the same variable x .

→ Add the coefficients: $5 + 1 = 6$.

→ Simplified total cost expression: $6x$.

Answer: $(5 + 1)x = 6x$

10. Combine like terms: $2x + 3x$

$$2x + 3x$$

→ Both terms contain the variable x — they are like terms.

→ Add the coefficients: $2 + 3 = 5$.

→ Simplified: $5x$.

Answer: $(2 + 3)x = 5x$

11. Simplify by combining like terms: $4x + 4x + 8 - 3$

$$4x + 4x$$

→ Combine x terms: $4x + 4x = 8x$.

→ Combine constants: $8 - 3 = 5$.

→ Simplified: $8x + 5$.

Answer: $(4 + 4)x = 8x$

12. A student writes $8x + 5x$ for the total cost of tickets. Simplify this expression.

$$8x + 5x$$

→ Both terms have the same variable x .

→ Add the coefficients: $8 + 5 = 13$.

→ Simplified total cost expression: $13x$.

Answer: $(8 + 5)x = 13x$



Solving one-step equations

13. Solve for x : $x + 6 = 18$

$$x + 6 = 18$$

→ Subtract 6 from both sides to isolate x .

$$\rightarrow x = 18 - 6 = 12.$$

→ Check: $12 + 6 = 18$. True.

Answer: $x = 18 - (6) = 12$

14. Solve for x : $2x = 7$

$$2x = 7$$

→ Divide both sides by 2 to isolate x .

$$\rightarrow x = 7 / 2 = 7/2.$$

Answer: $x = 7 \div 2 = 7/2$

15. After earning \$14 more, Maria has \$14 in all. Write and solve an equation to find how much she started with.

$$x + 14 = 14$$

→ Let x = starting amount. Equation: $x + 14 = 14$.

→ Subtract 14 from both sides: $x = 14 - 14 = 0$.

→ Maria started with \$0.

Answer: $x = 14 - (14) = 0$

16. Solve for x : $x + 4 = 19$

$$x + 4 = 19$$

→ Subtract 4 from both sides to isolate x .

$$\rightarrow x = 19 - 4 = 15.$$

→ Check: $15 + 4 = 19$. True.

Answer: $x = 19 - (4) = 15$

17. Solve for x : $2x = 5$

$$2x = 5$$

→ Divide both sides by 2 to isolate x .

$$\rightarrow x = 5 / 2 = 5/2.$$

Answer: $x = 5 \div 2 = 5/2$

18. After earning \$20 more, Maria has \$10 in all. Write and solve an equation to find how much she started with.

$$x + 20 = 10$$

→ Let x = starting amount. Equation: $x + 20 = 10$.

→ Subtract 20 from both sides: $x = 10 - 20 = -10$.

→ Maria started with \$-10.

Answer: $x = 10 - (20) = -10$



19. Solve for x: $x + 9 = 19$

$$x + 9 = 19$$

→ Subtract 9 from both sides to isolate x.

$$\rightarrow x = 19 - 9 = 10.$$

→ Check: $10 + 9 = 19$. True.

Answer: $x = 19 - (9) = 10$

20. Solve for x: $5x = 4$

$$5x = 4$$

→ Divide both sides by 5 to isolate x.

$$\rightarrow x = 4 / 5 = 4/5.$$

Answer: $x = 4 \div 5 = 4/5$

21. After earning \$16 more, Maria has \$28 in all. Write and solve an equation to find how much she started with.

$$x + 16 = 28$$

→ Let x = starting amount. Equation: $x + 16 = 28$.

→ Subtract 16 from both sides: $x = 28 - 16 = 12$.

→ Maria started with \$12.

Answer: $x = 28 - (16) = 12$



Evaluating expressions

22. Evaluate $6x + -2$ when $x = 5$.

$$6x - 2, x = 5$$

→ Substitute $x = 5$: $6(5) + -2 = 30 + -2 = 28$.

Answer: $6(5) + -2 = 30 + -2 = 28$

23. A plumber charges \$24 per hour plus a \$54 service fee. What is the total cost for 4 hours? Evaluate $24x + 54$ at $x = 4$.

$$24x + 54, x = 4$$

→ Total = $24(4) + 54 = 96 + 54 = \$150$.

Answer: $24(4) + 54 = 96 + 54 = 150$

24. Evaluate $4x^2 + 3$ when $x = 3$.

$$4x^2 + 3, x = 3$$

→ Substitute $x = 3$: $4(3)^2 + 3 = 4(9) + 3 = 36 + 3 = 39$.

Answer: $4(3)^2 + 3 = 36 + 3 = 39$

25. Evaluate $1x + 5$ when $x = 1$.

$$1x + 5, x = 1$$

→ Substitute $x = 1$: $1(1) + 5 = 1 + 5 = 6$.

Answer: $1(1) + 5 = 1 + 5 = 6$

26. A plumber charges \$57 per hour plus a \$35 service fee. What is the total cost for 2 hours? Evaluate $57x + 35$ at $x = 2$.

$$57x + 35, x = 2$$

→ Total = $57(2) + 35 = 114 + 35 = \149 .

Answer: $57(2) + 35 = 114 + 35 = 149$

27. Evaluate $2x^2 + -3$ when $x = 4$.

$$2x^2 + -3, x = 4$$

→ Substitute $x = 4$: $2(4)^2 + -3 = 2(16) + -3 = 32 + -3 = 29$.

Answer: $2(4)^2 + -3 = 32 + -3 = 29$

28. Evaluate $2x + 3$ when $x = 7$.

$$2x + 3, x = 7$$

→ Substitute $x = 7$: $2(7) + 3 = 14 + 3 = 17$.

Answer: $2(7) + 3 = 14 + 3 = 17$



29. A plumber charges \$24 per hour plus a \$24 service fee. What is the total cost for 4 hours? Evaluate $24x + 24$ at $x = 4$.

$$24x + 24, x = 4$$

$$\rightarrow \text{Total} = 24(4) + 24 = 96 + 24 = \$120.$$

Answer: $24(4) + 24 = 96 + 24 = 120$

30. Evaluate $4x^2 + -2$ when $x = 5$.

$$4x^2 + -2, x = 5$$

$$\rightarrow \text{Substitute } x = 5: 4(5)^2 + -2 = 4(25) + -2 = 100 + -2 = 98.$$

Answer: $4(5)^2 + -2 = 100 + -2 = 98$

